

SIMATSENGINEERING(SAVEETHASCHOOLOFENGINEERING)
DepartmentofComputerScienceandEngineering
ASSESSMENTTOOL 3–Hackathon

CourseCode:	DSA0405	CourseTitle:	FUNDAMENTALSOFDATA SCIENCE
COAssessed:	CO5–Hackathon	Assessment:	AssessmentTool3–Hackathon
Weightage:	40%of CIA	TotalMarks:	100
CO3Statement:	Analyzeandinterpretdatasetsusingdatapreprocessing,statisticalanalysis,visualization, and machine learning techniques for effective data-driven insights and decision-making.		
StudentName:	Kishore Kumar P	Reg.No.:	192424108
Section/Batch:	AI-DS	Date:	29/08/26

CodeofConduct

I Kishore Kumar P 192424108 certify that this submission is my original work and thatIhave adhered to the guidelinesspecifiedforthisassessment.Iunderstandthatanyviolationofacademicintegrityruleswillresultin disciplinary action.

SignatureoftheStudent: Kishore Kumar P

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Applicationof Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Failstoapply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Goodanalysis with logical reasoning	Basicanalysis with limited depth	Weak or no analysis;lacks reasoning	

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

DSA0405Fundamentals of Data Science

CO5-AT3 – Hackathon

Code:

```
loanguard-ai > backend > app > services >  ml_pipeline.py

5  import os
6  import json
7  import logging
8  import numpy as np
9  import pandas as pd
10 import joblib
11 from datetime import datetime
12 from typing import Dict, Any, Tuple, List, Optional
13
14 from sklearn.tree import DecisionTreeClassifier, export_text
15 from sklearn.model_selection import train_test_split, cross_val_score
16 from sklearn.metrics import (
17     accuracy_score, precision_score, recall_score,
18     f1_score, roc_auc_score, confusion_matrix, roc_curve,
19     classification_report,
20 )
21 from sklearn.preprocessing import LabelEncoder, StandardScaler
22 from sklearn.pipeline import Pipeline
23 from sklearn.compose import ColumnTransformer
24 from sklearn.preprocessing import OneHotEncoder
25
26 logger = logging.getLogger(__name__)
27
28 ARTIFACTS_DIR = os.path.join(os.path.dirname(os.path.dirname(os.path.dirname(__file__))), "artifacts")
29 DATA_DIR = os.path.join(os.path.dirname(os.path.dirname(os.path.dirname(__file__))), "data")
30
31
32 class MLPipeline:
33     """Complete ML pipeline for loan default prediction using CART."""
34
35     def __init__(self):
36         self.baseline_model: Optional[DecisionTreeClassifier] = None
37         self.pruned_model: Optional[DecisionTreeClassifier] = None
38         self.preprocessor: Optional[ColumnTransformer] = None
39         self.feature_names: List[str] = []
40         self.original_feature_names: List[str] = []
41         self.target_column: str = "Default"
```

Implementation:

SolarCalc

CAPSTONE

Simpson's 1/3 Rule & Romberg Integration

Home

Data Upload

Analysis

Mathematical Model

Numerical Methods

Comparison

Efficiency

Reports

Demo Data

Start Analysis

NUMERICAL METHODS • SOLAR ENERGY

Solar Collector Efficiency Analysis

Simpson's 1/3 Rule vs Romberg Integration

An interactive numerical analysis platform for processing solar data, estimating collected energy, comparing numerical integration methods, and evaluating solar collector performance.

Start Analysis

Explore Methodology

$O(h^4)$

Simpson Accuracy

Richardson

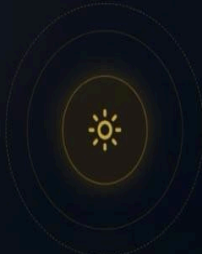
Romberg Extrap.

100%

Dynamic Calcs

SOLAR CORE SENSOR HUD

$\lambda: 280-2500\text{ nm}$



PEAK FLUX
~950 W/m²

INTEGRATION
 $O(h^4)$ / Richardson

LoanGuard AI

Overview

Risk Predictor

Explainable AI

Model Lab

Batch Scanner

Data Intelligence

About

Online

Run Prediction

ML ENGINEERING

Model Lab

Train, evaluate, compare, and optimize the CART risk model.

ALGORITHM
CART

CRITERION
Gini

PRUNING
CCP

TRAINING
30480 samples

TESTING
7621 samples

A SELECTED
0.00680

Baseline vs Pruned CART

METRIC	BASELINE	PRUNED	CHANGE
Accuracy	74.85%	84.88%	+ 10.03%
Precision	20.69%	0.00%	- 20.69%
Recall	23.44%	0.00%	- 23.44%
F1 Score	21.98%	0.00%	- 21.98%
ROC-AUC	53.72%	50.00%	- 3.72%
Tree Depth	44	0	-44
Leaves	4752	1	-4751

Baseline CART

Complex Tree

Depth: 44

Leaves: 4752

Higher Overfit Risk

Cost-Complexity Pruning

$\alpha = 0.00680$

Cross-validated across 50 candidates

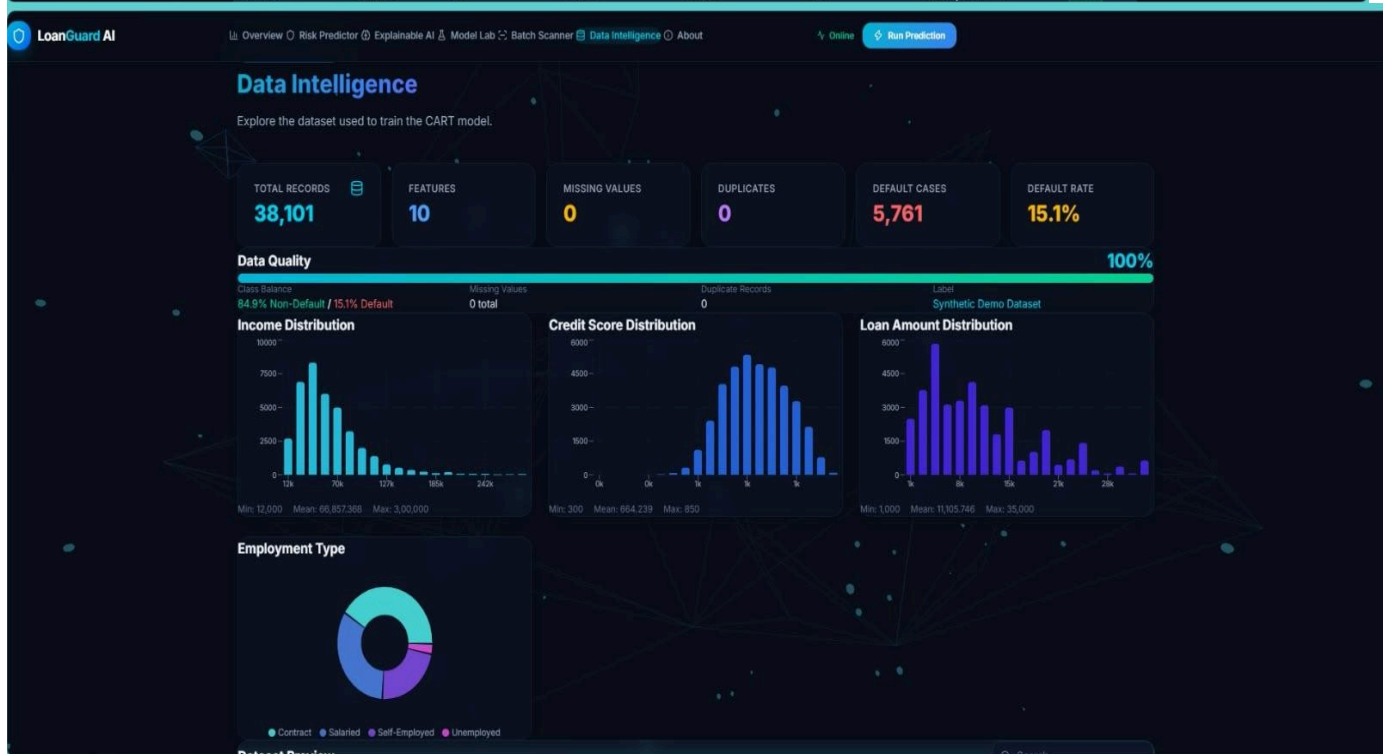
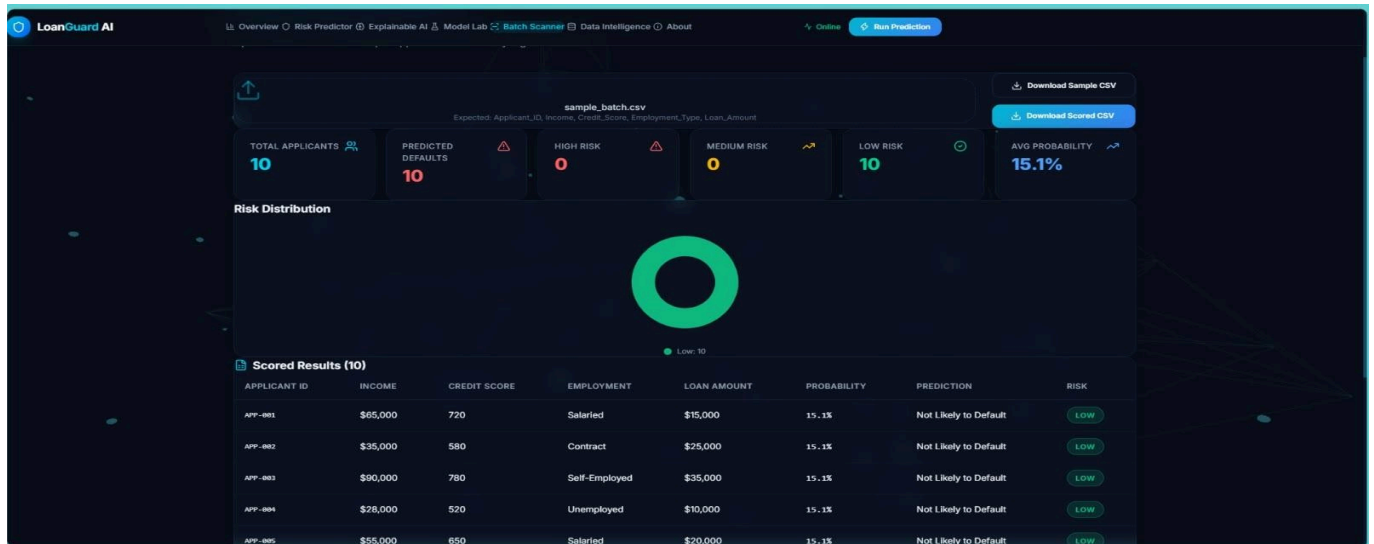
Pruned CART

Simplified Tree

Depth: 0

Leaves: 1

Better Generalization



Geotag:

